

Assignment 4

Aim - Write a program to compute longest common subsequence for the given strings of length atleast equal to 10.

```
#include <stdio.h>
#include <string.h>
```

```
# define MAX 100
```

```
void main ( ) {
    char a[MAX], b[MAX], LCS[MAX];
    int dp[MAX][MAX];
    int i, j, m, n, k;
```

```
    printf("Enter first string: ");
    scanf("%s", a);
    printf("Enter second string: ");
    scanf("%s", b);
```

```
    m = strlen(a);
    n = strlen(b);
    if (m < 10 || n < 10) {
        printf("Error: both strings must have
        atleast 10 characters.\n");
        return; return;
    }
```

```
    for (i = 0; i <= m; i++) {
        for (j = 0; j <= n; j++) {
```

```

    if (i == 0 || j == 0)
        dp[i][j] = 0;
    else if (a[i-1] == b[j-1])
        dp[i][j] = dp[i-1][j-1] + 1;
    else if (dp[i-1][j] > dp[i][j-1])
        dp[i][j] = dp[i-1][j];
    else
        dp[i][j] = dp[i][j-1];
}

```

```

}
printf("Length of LCS = %d\n",
        dp[m][n]);

```

```

i = m; j = n; k = dp[m][n];
lcs[k] = '\0';

```

```

while (i > 0 && j > 0) {
    if (a[i-1] == b[j-1]) {
        lcs[k-1] = a[i-1];
        i--; j--; k--;
    }
}

```

```

else if (dp[i-1][j] > dp[i][j-1])
    i--;

```

```

else

```

```

    j--;
}

```

```

printf("Longest Common Subsequence =
    %s\n", lcs);
}

```

Time Complexity: $O(m \times n)$
 Space Complexity: $O(m \times n)$

Enter first string: ASDFGHJKLQWERTYUIOP

Enter second string: ZXCVCBNMQWERTYUI

Length of LCS = 8

Longest Common Subsequence = QWERTYUI

Enter second string: POIUYTREWQZXCVCBNM

Length of LCS = 1

Longest Common Subsequence = P